Assignment 1 Summary Report: Constraint Mapping Across Data Sources

Applicant: Owen Mutch

Internship: Ecesis Investments 2026 Summer Power Systems Modeling Internship

Assignment: Constraint Mapping Across Data Sources

1. Objective

The objective of this assignment was to map power market constraints across three different data sources: PJMISO Market data, Dayzer, and Panorama. Each source describes constraints using different naming conventions and levels of detail, which creates a challenge when trying to identify whether multiple entries refer to the same underlying physical constraint.

The goal of my work was to create a consistent matching process that could identify likely corresponding constraints across the three datasets and produce a final CSV file with the required columns: market_constraint, dayzer_constraint, and pano_constraint.

2. Data Used

The assignment provided three constraint lists from different sources: PJMISO Market data, Dayzer, and Panorama. The Market dataset included separate fields for the constraint and contingency. The Panorama dataset also included separate monitored facility and contingency name fields. The Dayzer dataset provided a single constraint name field.

Because the sources used different structures, I created one comparable constraint string for each source. For the Market data, I combined the CONSTRAINT and CONTINGENCY fields. For Panorama, I combined the Monitored Facility and Contingency Name fields. For Dayzer, I used the NAME field as the main constraint description.

3. Methodology

My approach combined text normalization, fuzzy string matching, and confidence thresholds. First, I created a standardized constraint description for each source so the records could be compared more consistently. I then normalized the text by converting it to lowercase, removing punctuation and extra separators, standardizing voltage formatting, replacing common abbreviations such as xfmr with transformer, and removing extra spaces.

After cleaning the text, I used fuzzy string matching to compare each Market constraint against the Dayzer and Panorama constraint lists. I used the Market dataset as the base dataset because it had the fewest rows and because the final output required each row to begin with a market_constraint. For each Market constraint, the code selected the closest Dayzer match and the closest Panorama match based on a similarity score.

4. Handling Ambiguous Matches

Because different sources may describe the same physical constraint with different wording or levels of detail, I did not want to force every record into a match. After generating similarity scores, I classified each match into confidence categories: high confidence for scores of 90 or above, medium confidence for scores from 80 to 89, low confidence for scores from 70 to 79, and weak confidence for scores below 70.

For the final output, I kept high and medium confidence matches and blanked out low or weak matches. I chose this approach because an incorrect match could create a misleading view of congestion risk. A blank field is more transparent than forcing a poor match that may not represent the same underlying physical constraint.

5. Results

The final output file, matched_constraints.csv, contains 5,230 Market constraints and the best available corresponding Dayzer and Panorama matches. The required columns in the final CSV are market_constraint, dayzer_constraint, and pano_constraint.

In addition to the required output file, I created scored review files to support quality control. These include a file with match scores and confidence labels, a file containing questionable matches for manual review, and a summary statistics file showing the distribution of match confidence levels. This made it easier to evaluate how many matches were strong, moderate, or uncertain.

6. Insights and Conclusions

This assignment showed that constraint mapping is not just a simple text-matching problem. The same physical constraint can be described differently across systems, and the level of detail may vary by source. Market and Panorama data were easier to compare because both included separate facility and contingency information, while Dayzer required more reliance on the single name field.

The strongest matches generally occurred when the facility name, voltage level, and contingency description were all represented clearly across sources. Lower-confidence matches often appeared when one source used a more abbreviated name, when the contingency was described differently, or when one dataset contained less detail than another.

My final output takes a conservative approach by keeping stronger matches and leaving weaker matches blank. This creates a more reliable mapping file and avoids overstating certainty. For a trading desk, this is important because inaccurate constraint mapping could lead to an incorrect understanding of congestion risk.

7. Files Submitted

The main files included for this assignment are the Jupyter notebook showing the full process, the final matched constraints CSV file, and this summary report. I also included additional review files with match scores, confidence labels, questionable matches, and summary statistics to make the process more transparent.

The required final output file is matched_constraints.csv.